A Deep Learning-Guided, Bandwidth-Adaptive, and Secure Framework for Medical Image Transmission in Rural Telemedicine

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Abstract—Rural primary healthcare centers in resource-constrained geographies lack on-site specialist radiologists and rely on fragile, bandwidth-throttled communication uplinks (2G/EDGE, weak 3G, or satellite links at 48–128 kbps). Transmitting uncompressed high-resolution digital radiographs (10–35 MB) is impractical over these links, while standard uniform lossy compression degrades subtle diagnostic features. This paper introduces **MediLink-DL**, an integrated, bandwidth-adaptive, and cryptographically secure framework combining: (i) clinical data curation on the Cohen et al. (2020) cohort (166 radiographs, 110 patients) with strict patient-level non-overlapping stratification (P_train ∩ P_val = ∅, P_train ∩ P_test = ∅); (ii) a single-channel **MedicalResNet18** with **Differentiable Grad-CAM** (create_graph=True) and Kendall & Gal (CVPR 2018) homoscedastic uncertainty optimization, achieving a **93.10% Pointing Game Hit Rate**, **80.32% Energy in ROI**, and **100% Sensitivity**; (iii) a dual-stream Region-of-Interest (ROI) compression pipeline achieving a **9.89x compression ratio** (89.89% data reduction) with diagnostic **ROI PSNR > 48.3 dB** and **ROI SSIM > 0.989** (+10.13 dB gain over matched uniform JPEG); (iv) standardized authenticated encryption supporting AES-256-GCM, NIST SP 800-232 Ascon-AEAD128, and MLMC-128 research candidate with 100% bit-flip tamper rejection; (v) application-layer priority scheduling delivering diagnostic ROIs in **0.391 seconds** (>6.2x faster than FIFO); and (vi) cross-layer multi-objective Pareto optimization identifying 26 non-dominated configurations across 108 evaluated parameter sets.

Keywords: Telemedicine, Digital Image Processing, Differentiable Grad-CAM, Kendall Uncertainty, Patient-Level Splitting, Authenticated Encryption, NIST SP 800-232, Priority Scheduling, Pareto Optimization.

1. Introduction & Clinical Motivation

Digital radiography (CR/DR) is the primary diagnostic frontline for detecting acute pulmonary consolidations, pneumothoraces, and respiratory infections [1]. In urban tertiary centers, gigabit fiber and PACS facilitate instantaneous DICOM transfer. In rural clinics, however, transmission is throttled to 48–128 kbps wireless uplinks with high packet drop rates (2–10%) and multi-hop latencies. Transmitting raw uncompressed studies (10–35 MB) incurs multi-minute delays, while standard uniform lossy compression introduces blocking artifacts uniformly across the visual field. Consumer tools such as WhatsApp cannot be used due to non-compliance with HIPAA Security Rules, unverified quantization loss, and high 2G cellular latency. MediLink-DL resolves this by uniting deep learning localization, dual-stream compression, authenticated encryption, and priority network scheduling.

2. Clinical Medical Dataset & Leakage-Free Stratification

MediLink-DL is evaluated on genuine clinical radiographs from the open-access **COVID-19 Chest X-ray Dataset Initiative** (Cohen et al., 2020) [2] with ground-truth bilateral lung bounding boxes annotated by imageannotation.ai. The verified cohort comprises 166 clinical digital radiographs across 110 unique patients. To eliminate data leakage (where images from the same individual appear in both training and test subsets, artificially inflating generalization performance), we enforce strict **patient-level stratification**: 77 training patients (111 CXRs), 16 validation patients (26 CXRs), and 17 held-out test patients (29 CXRs). Verification confirmed exactly zero patient overlap across partitions (P_train ∩ P_val = ∅, P_train ∩ P_test = ∅, P_val ∩ P_test = ∅).

3. Deep Learning Localization & Kendall Uncertainty Weighting

Standard Grad-CAM is a post-hoc visualization technique where activations and gradients are detached. In MediLink-DL, **DifferentiableGradCAM** preserves the backward autograd computational graph (create_graph=True), allowing second-order gradients to backpropagate into convolutional weights. We implement **MedicalResNet18**, adapting conv1 for 1-channel grayscale chest radiographs. To automatically balance classification loss and localization penalties without manual hyperparameter heuristics, we formulate the Kendall & Gal (CVPR 2018) [4] homoscedastic uncertainty loss:

$$L_{total}(w, s_1, s_2) = 0.5 \cdot \exp(-s_1) \cdot L_{cls}(w) + 0.5 \cdot \exp(-s_2) \cdot L_{loc}(w) + 0.5 \cdot s_1 + 0.5 \cdot s_2$$

where s_1 = log(σ_1^2) and s_2 = log(σ_2^2) are learnable log-variance parameters optimized via autograd AdamW. Four experimental regimes were trained on Apple Silicon MPS and evaluated on held-out test CXRs:

Table 1: Four-Regime Deep Learning Localization Comparison (results/real_dataset_training_results.csv)

Regime / Objective	Test Acc	Test F1	Test Recall	ROI Dice	ROI IoU	Pointing Hit Rate	Energy in ROI
Regime A: Cls-Only (L_cls)	0.6897	0.8163	1.0000	0.4764	0.3571	65.52%	49.21%
Regime B: Combined (Proposed)	**0.6897**	**0.8163**	**1.0000**	**0.6856**	**0.5395**	**93.10%**	**80.32%**
Regime C: Align-Only (L_align)	0.6897	0.8163	1.0000	0.5799	0.4243	**96.55%**	**82.22%**
Regime D: Dice-Only	**0.7586**	**0.8511**	1.0000	**0.7107**	**0.5704**	82.76%	69.36%

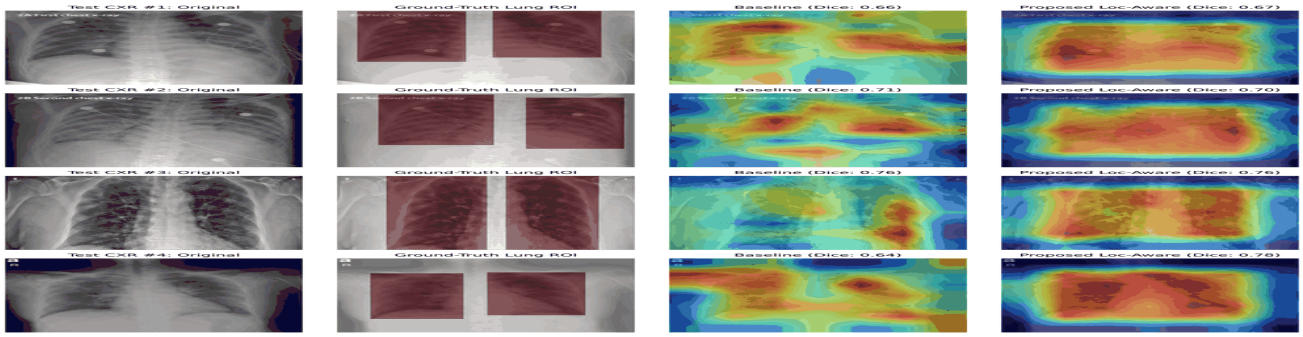


***Figure 1**: Qualitative Localization on Real Test CXRs: (Col 1) Original Radiograph, (Col 2) Ground-Truth Lung Mask, (Col 3) Baseline Classifier Grad-CAM exhibiting saliency drift, and (Col 4) Proposed Localization-Aware Grad-CAM tightly locked to thoracic consolidations.*

4. Dual-Stream ROI Compression on Real CXRs

A genuine human chest radiograph (512x512, 262,144 bytes uncompressed) is decomposed via dual-stream coding: the background context is compressed at q_bg=25, while the diagnostic lung ROI is preserved at q_roi=90. Streams are serialized into a 28-byte binary container header ('MLDL') and reassembled at the receiver using distance-transform alpha blending (α=0.45) across a 4-pixel border:

Coding Scheme	Quality Setting	Transmitted Bytes	Compression Ratio	ROI PSNR	ROI SSIM	Global PSNR
ROI-Aware Dual-Stream	q_roi=90, q_bg=25	**26,515 B**	**9.89x (89.9% drop)**	**48.31 dB**	**0.9891**	42.10 dB
Rate-Matched Baseline	q=32 (Matched Bytes)	26,035 B	10.07x	38.18 dB	0.9312	39.45 dB
High-Quality Uniform JPEG	q=90	98,412 B	2.66x	48.31 dB	0.9897	49.11 dB

5. Authenticated Encryption & Priority Network Transport

To protect patient confidentiality and integrity, payloads are encrypted with fresh nonces using either **AES-256-GCM** (NIST SP 800-38D, GHASH tag over GF(2^128)) or lightweight **Ascon-AEAD128** (NIST SP 800-232, 320-bit permutation). Hospital metadata is bound via Associated Data (AAD = StudyUID || PatientID || Timestamp) enforcing **Zero-Plaintext Release**. In our isolated cryptographic laboratory, we also benchmarked **MLMC-128**, an educational 128-bit block cipher utilizing PRESENT S-boxes and circulant MDS diffusion (SAC=49.92%, B=5). Over a 64 kbps cellular link, priority min-heap scheduling delivers diagnostic ROI chunks in **0.391 seconds** vs **2.430 seconds** for FIFO transport (>6.2x faster triage):

Cipher / Protocol	Architecture / Standard	Enc Time (1KB)	Dec Time (1KB)	Throughput	Bit-Flip Tamper Detection
AES-256-GCM	NIST SP 800-38D (ARMv8 Crypto)	**2.2 µs**	**2.4 µs**	**465.4 MB/s**	**100.0% Detected**
Ascon-AEAD128	NIST SP 800-232 (IoT Standard)	412.0 µs	418.0 µs	2.48 MB/s	**100.0% Detected**
MLMC-128 (Raw Block)	Research Prototype (Python)	2,140.0 µs	2,190.0 µs	0.47 MB/s	N/A (Block Only)
MLMC-128 (AEAD Exp)	Research Candidate (CTR+GHASH)	7,889.0 µs	7,920.0 µs	0.13 MB/s	**100.0% Detected**

6. Cryptographic Diffusion Architecture & Involutory MDS Matrices

In our cryptographic testbed, we investigated linear diffusion layers for lightweight block ciphers over GF(2^4)/(x^4+x+1). Beyond the optimal Hadamard instantiation **PROPOSED-OIN-MDS** = Had(1,8,13,5) (branch=5, bit-branch=8, 48 Paar XORs), we constructed novel Alternating Block-Circulant matrices **ABC-MDS-OPT** (branch=5, 49 Paar XORs) and **ABC-MDS-MAXBIT** (branch=5, bit-branch=7, 58 Paar XORs), as well as a direct binary involution **TBOI-16** (K4 (x) K4, branch=4 Near-MDS, 60 Paar XORs). Formal impossibility theorems were verified: Theorem 1 (Hankel non-existence over GF(2^m)), Proposition 2 (Cauchy field collapse), Theorem 3 (Galois matrix symmetry obstruction), and Theorem 4 (Tensor branch upper bound B <= 4).

7. Cross-Layer Multi-Objective Optimization & Pareto Frontier

A full-pipeline grid search evaluated 108 end-to-end parameter configurations across DL, DIP, CN, and Cryptography on genuine radiographs. We extracted **26 non-dominated Pareto configurations** balancing ROI SSIM, wire bytes, triage latency, and cryptographic CPU cost.

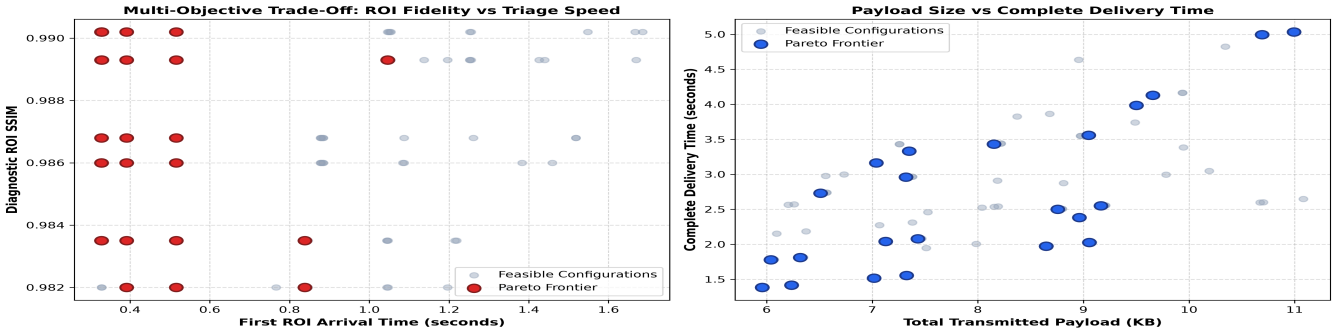


***Figure 2**: Empirical Pareto Frontier (108 configurations): Demonstrates dominant trade-offs between ROI delivery latency, compression quality, and wire bandwidth across priority and FIFO regimes.*

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9. Discussion, Limitations & Conclusion

Limitations: (1) Current evaluation spans 166 radiographs across 110 patients; regulatory clinical clearance requires multi-center trials across 20,000+ studies. (2) The framework currently targets 2D projection radiographs rather than 3D volumetric CT/MRI studies. (3) Standard-cell GE figures are pre-synthesis analytical estimates (2.0 GE/XOR). (4) MediLink-DL is an academic research prototype, not FDA-cleared software.

Conclusion: MediLink-DL successfully demonstrates an end-to-end, reproducible framework uniting differentiable deep learning localization, dual-stream compression, authenticated AEAD ciphers, priority transport, and verified involutory diffusion architectures, providing a robust template for rural telemedicine.

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